

Literal Re-translation as a Method for AI Text Disguise and Detection Evasion

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Voight Kampff Solution Approach

- My core method involved back translation (EN $\rightarrow$... $\rightarrow$ EN) but with additional prompting encouraging literal translation.
- Tested 4 Indian languages (Hindi, Gujarati, Tamil, Bengali) and 2 European languages (Spanish, French)
- Locally validated results using GPT-Zero, and vibe-reading.
- Primarily used Claude Sonnet 3.5
- Used 2 versions of the prompt:
 - Basic prompt instructing for literal retranslation
 - Advanced prompt instructing literal retranslation being more adversarial, added explicit instructions for the model to mimic human flaws.

Findings

- Local testing across languages showed inability of LLM to follow adversarial instructions for relatively richer Indian languages (Gujarati, Tamil, Bengali) leading to the final output being still AI - like
- European languages when used in retranslation, showed good adversarial instruction following but the final result still followed the common AI phrases and words (“Thrilled”)
- Hindi was a sweet spot, with good adversarial instruction following and not using AI-ish English phrases/format

Results

- Basic prompt with hindi retranslation was relatively unsuccessful in fooling the PAN AI detection models, getting the final Brier score of 0.72224 and C@1 score of 0.66560
- Adversarial prompt with hindi retranslation was more successful and got Brier - 0.42498 C@1 - 0.34352